IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received 29 December 2024, accepted 11 February 2025, date of publication 11 March 2025, date of current version 4 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3549466

RESEARCH ARTICLE

SMPL Variable Model for 3D Reconstruction and Image Fusion in Animation Media Applications

TIAN XIE[1] AND KUNPENG ZHAO[2]
[1]School of Design and Art, Xijing University, Xi'an 710123, China
[2]Geely University, Chengdu 641423, China
Corresponding author: Tian Xie (m15902901770@163.com)

ABSTRACT 3D human reconstruction is a pivotal research area in computer vision and computer graphics, with broad applications in fields such as virtual reality, animation media, and medical simulation. Traditional reconstruction approaches often rely on fixed-shape models, which limits their ability to accurately represent the dynamic soft tissue variations and complex interactions with the surrounding environment. These methods face challenges in managing occlusions in multi-person scenes, capturing a diverse range of poses, and handling extensive deformations in real-world scenarios, which constrains their practical application. To overcome these limitations, this paper introduces an enhanced 3D reconstruction model using the SMPL variable model, which integrates a soft tissue layer, UV space mapping, and a force-conditioning mechanism. The proposed model incorporates a variable-thickness soft tissue layer to better capture anatomical variations, utilizes UV space mapping to streamline deformation calculations, and employs force-conditioning to realistically simulate intricate interactions. Experimental results on datasets such as ShapeNet, ShapeNetCore, ABC, and Scan2CAD demonstrate that our approach surpasses current state-of-the-art methods in terms of accuracy, recall, F1 score, and AUC, while also achieving greater efficiency in both training and inference phases. This research not only delivers a more accurate and computationally efficient solution for 3D reconstruction but also contributes novel insights into multi-person interaction modeling and enhances generalization in real-world animation media applications.

INDEX TERMS 3D reconstruction, feature extraction, image quality improvement, image fusion, robot vision.

I. INTRODUCTION

The task of 3D reconstruction of animation media images has gained immense importance in recent years due to its wide applicability in fields such as virtual reality, gaming, film production, and medical simulation [1]. Accurate reconstruction not only enhances the realism and interactivity of digital environments but also allows for sophisticated analysis of human motion, posture, and interactions in virtual spaces [2]. The increasing demand for high-quality animation and media content has led to a growing interest in developing robust and efficient 3D reconstruction methods [3]. Traditional methods face challenges in capturing the complexity of human forms, including diverse clothing, dynamic soft tissue deformations, and intricate interactions with the environment [4]. Additionally, the proliferation of multi-person interactions in animation and media scenes necessitates models that can handle occlusions and pose diversity effectively [5]. Hence, the development of advanced 3D reconstruction methods that are both computationally efficient and capable of producing highly detailed and accurate reconstructions is not only desired but essential. The evolution of this field can be segmented into three distinct phases: traditional symbolic AI and knowledge representation, data-driven machine learning methods, and deep learning with pre-trained models [6].

To address the initial challenges of 3D reconstruction, early methods predominantly relied on symbolic AI and knowledge representation [7]. These traditional methods were based on

The associate editor coordinating the review of this manuscript and approving it for publication was Alba Amato.

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

handcrafted geometric models, parametric representations, and physical simulation rules to generate 3D forms. One of the most notable examples is the Skinned Multi-Person Linear (SMPL) model, which uses a set of parameters to capture the human body's shape and pose [8]. By integrating anatomical knowledge and physical constraints, these models provided a structured approach to reconstructing 3D shapes. SCAPE and other physics-based models introduced further refinements by simulating skin deformation and muscle movements [9]. However, despite their structured nature, these symbolic AI methods faced significant drawbacks [10]. The fixed topology of parametric models was inadequate for capturing the dynamic soft tissue deformations and the nuances of complex clothing. Moreover, the reliance on manual parameter tuning and predefined rules limited their flexibility and scalability [11]. The inability to generalize to a wide variety of poses, shapes, and interactions meant that these methods were constrained when applied to the diverse and dynamic nature of animation media images. Recognizing these limitations, researchers began exploring more automated and flexible data-driven approaches [12].

In response to the constraints of symbolic AI-based methods, the next phase saw a shift towards data-driven machine learning approaches. Researchers leveraged large-scale datasets and statistical techniques to learn more adaptable models of human shape and motion [12]. Techniques such as Principal Component Analysis (PCA), Gaussian Process Models, and statistical shape modeling were employed to capture a broad variety of body shapes and poses from data. Multi-view stereo and photogrammetry techniques also gained prominence, allowing for automatic reconstruction of 3D models from 2D image inputs [13]. These approaches offered a significant step forward by reducing reliance on handcrafted rules and enabling models to learn from data, thereby improving generalization to some extent [13]. However, these data-driven methods still exhibited limitations. They often required extensive data pre-processing, and their performance heavily depended on the diversity and quality of the training datasets [14]. Additionally, they struggled with generalizing to new or unseen poses and interaction scenarios that were not well-represented in the training data. The computational cost of optimizing shape parameters and refining reconstructions also posed challenges for real-time applications. To overcome these issues, the field began to incorporate machine learning models, setting the stage for the next evolution in deep learning-based methods [15].

Building on the foundation of data-driven models, the advent of deep learning and pre-trained models marked a significant milestone in 3D reconstruction. Deep learning techniques, especially Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), have enabled more complex mappings from 2D images to 3D shapes. Methods like DensePose, HMR, and SPIN harness large-scale datasets and the powerful feature extraction capabilities of deep neural networks to predict human body meshes directly from single images or videos [16]. Pre-trained models, in particular, facilitated transfer learning, allowing the application of learned features to various tasks and datasets, thereby enhancing the models' generalization capabilities. Despite these advancements, deep learning-based approaches are not without limitations. They often require vast amounts of labeled data for training, which can be difficult and time-consuming to collect [17]. Additionally, they are computationally expensive, demanding high-end hardware resources for both training and inference [18]. Moreover, while these models can generate realistic static reconstructions, they sometimes fall short in accurately simulating dynamic soft tissue deformations and handling complex interactions in multi-person scenes. These challenges indicate that there is room for further innovation in developing more efficient and versatile methods for 3D reconstruction of animation media images [19].

In light of the aforementioned limitations, we introduce our novel method, AniReconstructNet, which integrates elements from traditional knowledge-based modeling, data-driven approaches, and deep learning techniques to tackle the complex challenges of 3D reconstruction. AniReconstructNet employs a variable-thickness soft tissue layer to simulate realistic deformations, utilizes UV space mapping for computational efficiency, and implements a force-conditioning mechanism to manage dynamic interactions in multi-person scenes. The proposed method offers several advantages:

- Introduces a variable-thickness soft tissue layer and force-conditioning mechanism, enabling the model to capture intricate soft tissue deformations and dynamic interactions that previous methods struggled with.
- Leverages UV space mapping to simplify computations, resulting in a model that is not only efficient in multi-scene and multi-person environments but also highly generalizable across various poses and interactions.
- Achieves superior performance in accuracy, recall, F1 score, and AUC on multiple benchmark datasets, significantly outperforming existing state-of-the-art methods in both reconstruction quality and computational efficiency.

II. RELATED WORK

A. TRADITIONAL PARAMETRIC HUMAN MODELS

Traditional 3D human reconstruction methods often rely on parametric models, such as SCAPE and SMPL, which use a fixed set of shape and pose parameters to represent the human body. These models are built on statistical shape modeling, where a large set of human scans is used to learn a linear deformation space that can represent various human shapes and poses [20]. The Skinned Multi-Person Linear (SMPL) model is one of the most popular parametric models, as it provides a compact and efficient representation of the human body by using a small number of parameters to capture both global pose changes and local shape variations [21]. However, despite their success, these parametric models have several limitations. Firstly, they assume a fixed topology and cannot model dynamic soft tissue deformations, which are

crucial for realistic human body simulation, especially in complex poses or interactions [22]. Secondly, these models struggle to handle real-world scenarios where occlusions, diverse clothing, and multiple interacting bodies are present. To address these limitations, recent works have extended the basic SMPL model by adding soft tissue layers or clothing deformations, yet they still fall short in accurately capturing the intricate interactions between human bodies and the environment. Our approach builds upon the SMPL model by introducing a variable-thickness soft tissue layer and force-conditioning mechanism to simulate more realistic interactions, providing a more flexible framework for dynamic 3D reconstruction [23].

B. DATA-DRIVEN DEEP LEARNING METHODS

With the rise of deep learning, data-driven methods have become the dominant approach for 3D human reconstruction. These methods utilize neural networks, particularly convolutional neural networks (CNNs) and generative adversarial networks (GANs), to directly learn complex mappings from input data (such as images, videos, or point clouds) to 3D shapes [24]. Approaches like DensePose, HMR, and SPIN leverage large-scale datasets to train models that can predict 3D body meshes from 2D images. Unlike parametric models, deep learning-based methods can handle diverse input types, such as monocular images, multi-view images, and depth scans, making them more versatile for various applications [25]. However, deep learning methods face challenges in generalizing to new poses, shapes, and interaction scenarios that are not well-represented in the training data. Additionally, most data-driven methods primarily focus on static reconstruction, lacking the ability to simulate dynamic soft tissue deformations and interactions [26]. Recent methods have attempted to incorporate physics-based modeling into neural networks, but they often involve high computational costs and complex training processes. Our proposed method addresses these issues by introducing UV space mapping, which simplifies the deformation process, and force-conditioning, which allows for better handling of interactions, leading to more accurate and efficient 3D reconstructions.

C. INTERACTION-AWARE 3D RECONSTRUCTION

Interaction-aware 3D reconstruction is an emerging research direction focusing on accurately modeling how human bodies interact with objects and other people. Previous works, such as ContactOpt and PROX, use contact modeling to enforce constraints on human-object interactions [27]. These methods often employ optimization techniques to refine body poses based on the physical properties of the environment, such as contact points, friction, and surface normal alignment [23]. While effective in specific scenarios, these methods are typically limited by their dependence on explicit contact modeling and predefined physical rules, making them less flexible in complex, dynamic scenes. More recent research has explored learning-based approaches to capture interactions, such as using neural networks to predict contact maps or force fields [28]. However, these methods often require extensive labeled data and struggle to generalize to unseen interaction scenarios, especially in multi-person environments. Our approach advances this field by introducing a force-conditioning mechanism that directly models the influence of external forces in the UV space, enabling the simulation of various interactions without the need for explicit contact modeling. Additionally, our model can handle multi-person interactions through independent UV mappings for each individual, allowing it to adapt to complex and dynamic scenes more effectively.

III. METHODOLOGY

A. OVERVIEW

In this work, we propose a novel approach for the 3D reconstruction of human bodies and animation media images by building upon the SMPL model, and enhancing it to capture more complex deformations in realistic scenes. The method leverages the foundational principles of SMPL and extends them through the integration of additional modules tailored for more diverse and nuanced soft tissue modeling and geometric complexity of the human form, such as clothing and dynamic poses. Our method introduces a comprehensive framework that allows for efficient reconstruction, detailed soft-tissue simulation, and generalization to various input types, including multi-person interactions and complex occlusions. The flow of our model can be broken down into several key stages. First, a basic human body mesh is generated using the core SMPL model. This mesh is then enriched through the incorporation of a soft tissue layer, allowing for realistic deformations in response to external forces or interactions. Further refinements include the mapping of these deformations onto 2D UV coordinates, where a neural network processes the data for efficient simulation and reconstruction. Finally, these results are mapped back to a 3D form, producing the reconstructed mesh with intricate details.

In this paper, we organize our method's explanation into three subsections: *Preliminaries*, presented in Section III-B, which introduces the fundamental mathematical framework of our approach; *Dynamic Soft Tissue Modeling*, discussed in Section III-C, which describes the core enhancements made to the SMPL model to simulate soft tissue; and *Interaction and Generalization*, outlined in Section III-D, which elaborates on the strategies used to handle complex interactions and occlusions in multi-person scenes. This structure allows for a systematic exploration of the technical innovations, ultimately leading to a model capable of real-time and scalable 3D reconstruction from limited input data.

B. PRELIMINARIES

In this section, we formalize the problem of 3D reconstruction and deformation simulation within the context of animation

media images and human body modeling. Let us represent the 3D human body shape by a mesh $\mathcal{M}$, which consists of a set of vertices $\mathcal{V} \in \mathbb{R}^{N\times 3}$ and faces $\mathcal{F}$. The Skinned Multi-Person Linear (SMPL) model parameterizes $\mathcal{M}$ using a set of shape parameters $\beta \in \mathbb{R}^{|\beta|}$ and pose parameters $\theta \in \mathbb{R}^{|\theta|}$. The SMPL model generates the 3D mesh $\mathcal{M}(\beta, \theta)$, encapsulating the variations in body shapes and joint positions in response to changes in β and θ.

The core challenge in human body reconstruction involves not only generating a geometrically accurate model $\mathcal{M}$ but also ensuring that the model can adapt to various external interactions, dynamic movements, and clothing deformations. To address this, we introduce an additional soft tissue layer $\mathcal{T}$, which is represented as an offset field $\delta\mathcal{V}$ over the baseline mesh vertices, i.e., $\mathcal{V}_{\text{final}} = \mathcal{V} + \delta\mathcal{V}$. The soft tissue layer is characterized by a thickness parameter t and a deformation mapping function $\mathcal{D}(\theta, F)$, where F denotes external forces acting on the body surface. The goal is to model $\delta\mathcal{V}$ as a function of θ, β, and F, thus allowing $\mathcal{T}$ to simulate realistic interactions and deformations.

To further generalize the representation of external interactions, we map the 3D shape $\mathcal{M}$ and the soft tissue deformations $\delta\mathcal{V}$ onto a 2D UV coordinate system using the parameterization $\phi : \mathcal{V} \to \mathcal{U} \in \mathbb{R}^{512\times 512}$. This 2D mapping simplifies the computation of deformations and allows the application of 2D convolutional neural networks (CNNs) for fast and efficient inference of the deformed shapes. Specifically, we define a UNET-based neural network $\mathcal{N}$ that processes the UV-mapped inputs to predict $\delta\mathcal{U}$, the deformation field in the UV space. The predicted deformation $\delta\mathcal{U}$ is then transformed back into the 3D space to update the mesh vertices $\mathcal{V}$.

Mathematically, given a set of input parameters $\{\beta, \theta, F\}$, the model seeks to solve:

$$\mathcal{V}_{\text{final}} = \mathcal{V} + \phi^{-1}(\delta\mathcal{U}(\phi(\mathcal{V}), \theta, F)), \tag{1}$$

where ϕ^{-1} denotes the inverse UV mapping operation. The deformation field $\delta\mathcal{U}$ is obtained through the neural network $\mathcal{N}$:

$$\delta\mathcal{U} = \mathcal{N}(\phi(\mathcal{V}), \theta, F). \tag{2}$$

The representation of forces F in the UV space is crucial for modeling various interactions. We define F as a set of force vectors $\{F_i\}$ applied at different vertices in the UV space, characterized by their magnitudes and directions. The force field F is parameterized as:

$$F = \{(u_i, v_i, f_i) | (u_i, v_i) \in \mathcal{U}, f_i \in \mathbb{R}^3\}, \tag{3}$$

where (u_i, v_i) are the UV coordinates and f_i is the force vector at the corresponding vertex. This formulation allows the network to learn complex force-deformation relationships and predict the soft tissue's response to various interactions.

The preliminaries set the mathematical foundation for our model, emphasizing the extension of SMPL with a dynamic soft tissue layer and the mapping of 3D deformations onto a 2D UV space. This approach facilitates efficient simulation and reconstruction of complex body deformations under diverse interactions, forming the basis for the modules discussed in the subsequent sections.

C. DYNAMIC SOFT TISSUE MODELING

In this section, we introduce our approach to dynamic soft tissue modeling, which extends the SMPL framework by integrating an additional layer for simulating realistic deformations. The objective of this module is to handle interactions with external objects, simulate soft tissue dynamics, and adapt to diverse body shapes and poses. Our approach introduces several key innovations to enhance the accuracy and generalizability of 3D reconstruction.

Variable-Thickness Soft Tissue Layer: The soft tissue layer $\mathcal{T}$ is a critical component in accurately simulating the dynamic deformations of the human body during interactions and movements. In our model, $\mathcal{T}$ is represented as a thickened mesh surface with a spatially varying thickness across different body regions. This approach allows the model to capture complex, non-uniform deformations that occur in various poses and under external forces, providing a more flexible representation compared to the basic SMPL model.

The thickness of the soft tissue layer is parameterized by a set of values $\mathbf{t} = \{t_i\}_{i=1}^N$, where t_i denotes the thickness at the i-th vertex of the base mesh $\mathcal{M}$. This parameterization introduces a continuous variability in thickness across the surface, facilitating the modeling of anatomically accurate deformations. The base mesh $\mathcal{M}$ consists of a set of vertices $\mathcal{V} \in \mathbb{R}^{N\times 3}$, and the final deformed mesh $\mathcal{V}_{\text{final}}$ is expressed as:

$$\mathcal{V}_{\text{final}} = \mathcal{V} + \delta\mathcal{V}, \tag{4}$$

where $\delta\mathcal{V} = \{\delta v_i\}_{i=1}^N$ represents the offset field indicating the displacement of each vertex due to soft tissue deformations. Each offset $\delta v_i \in \mathbb{R}^3$ is a vector that accounts for the changes in the soft tissue at vertex i based on the local thickness and the external forces applied.

The deformation of this soft tissue layer is driven by external forces $\mathbf{F} = \{F_i\}_{i=1}^N$, where each force vector $F_i \in \mathbb{R}^3$ represents the influence of interactions, gravity, and other environmental factors at vertex i. To model the influence of external forces on the soft tissue, we introduce a deformation mapping function $\mathcal{D}(\theta, F, \mathbf{t})$ that computes the offset δv_i as a function of the body's pose θ, the external force F_i, and the local tissue thickness t_i. The mapping is defined as:

$$\delta v_i = \mathcal{D}(\theta, F_i, t_i) = t_i \cdot \mathbf{R}(\theta) \cdot F_i, \tag{5}$$

where $\mathbf{R}(\theta)$ is a rotation matrix dependent on the pose parameters θ, capturing how the body orientation affects the direction and magnitude of the deformation. The multiplication by the thickness t_i ensures that areas with a larger thickness exhibit more significant deformations under the same force, allowing for anatomically accurate simulations of soft tissue behavior.

To further enhance the realism of deformations, we incorporate a non-linear stiffness function $\mathcal{S}(t_i)$, which modulates the response of the soft tissue based on its thickness. Thicker areas, such as the torso, have higher stiffness and resist deformation, while thinner regions, like the arms and legs, are more flexible. The stiffness function is defined as:

$$\mathcal{S}(t_i) = \alpha \exp(-\beta t_i), \tag{6}$$

where α and β are empirically determined constants that control the stiffness variation. The deformation mapping function is then adjusted to include the stiffness modulation:

$$\delta v_i = \mathcal{S}(t_i) \cdot t_i \cdot \mathbf{R}(\theta) \cdot F_i. \tag{7}$$

In our implementation, the deformation offsets $\delta\mathcal{V}$ are computed in a hierarchical manner, starting from coarse to fine details. The base mesh $\mathcal{M}$ undergoes an initial global deformation influenced by the overall body pose θ, followed by localized deformations that account for soft tissue dynamics. This hierarchical approach allows the model to capture both large-scale movements and fine-grained deformations, ensuring a realistic simulation of human body dynamics.

The soft tissue layer's influence extends to the interaction with clothing and external objects. During interactions, collision forces F_i^{col} are generated based on contact points between the body surface and external entities. These collision forces are incorporated into the total force vector:

$$F_i = F_i^{\text{ext}} + F_i^{\text{col}}, \tag{8}$$

where F_i^{ext} represents other external forces such as gravity. The deformation mapping function $\mathcal{D}(\theta, F, \mathbf{t})$ then calculates the corresponding offsets, allowing the model to simulate realistic responses to interactions, such as clothing stretching over the body or compressing under an external load. The variable-thickness soft tissue layer provides a more anatomically accurate and flexible model for human body deformations. By incorporating the thickness-dependent stiffness modulation and interaction forces, the model extends beyond the limitations of fixed-topology parametric models, enabling the simulation of complex, realistic deformations that occur during diverse interactions and movements.

UV Space Mapping for Efficient Deformation Simulation: To efficiently simulate complex deformations of the human body, we employ a UV space mapping strategy that projects the 3D body mesh, soft tissue properties, and external forces onto a 2D UV coordinate system. This approach simplifies the computation of deformations, allowing for the application of convolutional neural networks (CNNs) in UV space to process high-resolution inputs more efficiently. The UV space serves as a flattened representation of the 3D body surface, preserving both the spatial relationships and local geometric properties of the mesh.(As shown in Figure 1)

Let $\phi : \mathcal{V} \to \mathcal{U} \in \mathbb{R}^{512\times512}$ be the UV parameterization function that maps each vertex v_i of the 3D mesh $\mathcal{V}$ to its corresponding 2D UV coordinates (u_i, v_i). This UV mapping process also projects other properties, such as the soft tissue thickness $\mathbf{t}$ and external forces $\mathbf{F}$, into the 2D UV space:

$$\phi(\mathbf{t}) = \{t_i \mid (u_i, v_i) = \phi(v_i)\}, \tag{9}$$

$$\phi(\mathbf{F}) = \{F_i \mid (u_i, v_i) = \phi(v_i)\}, \tag{10}$$

where $\phi(\mathbf{t})$ and $\phi(\mathbf{F})$ denote the UV-mapped thickness and external forces, respectively. By mapping these properties into the UV space, we obtain a unified, fixed-size 2D representation that can be processed using deep learning models designed for image-based inputs.

Using this parameterization, we design a UNET-based neural network $\mathcal{N}$ that takes the UV-mapped mesh, thickness, and force fields as inputs. Specifically, the input to the network is a set of channels containing the UV-mapped mesh coordinates $\phi(\mathcal{V})$, the thickness map $\phi(\mathbf{t})$, and the external force map $\phi(\mathbf{F})$. Additionally, the pose parameters θ are encoded into a latent vector $\mathbf{z}_\theta$ and concatenated with the input features in the UV space. The network then predicts the deformation field $\delta\mathcal{U}$, which represents the displacements in UV space:

$$\delta\mathcal{U} = \mathcal{N}(\phi(\mathcal{V}), \phi(\mathbf{t}), \phi(\mathbf{F}), \mathbf{z}_\theta). \tag{11}$$

The UNET architecture is particularly well-suited for this task due to its ability to capture both local details and global context through its encoder-decoder structure and skip connections. In the UV space, local deformations are influenced by neighboring vertices, while global features, such as overall body posture and external forces, are incorporated via the latent pose vector $\mathbf{z}_\theta$. This results in a deformation field $\delta\mathcal{U} \in \mathbb{R}^{512\times512\times3}$, where each UV coordinate (u_i, v_i) corresponds to a displacement vector in 3D space.

Once the deformation field $\delta\mathcal{U}$ is predicted in the UV space, it must be transformed back to the 3D mesh. The inverse mapping $\phi^{-1} : \mathcal{U} \to \mathcal{V}$ is applied to map the UV displacements back to the original 3D vertices, updating their positions:

$$\mathcal{V}_{\text{final}} = \mathcal{V} + \phi^{-1}(\delta\mathcal{U}), \tag{12}$$

where $\phi^{-1}(\delta\mathcal{U})$ represents the UV-mapped deformation applied to the original 3D mesh vertices. This inverse mapping ensures that the displacements predicted in UV space correspond accurately to the surface geometry of the 3D mesh.

To handle interactions in the UV space, we introduce a force-conditioning mechanism. During training, the network learns to interpret how the UV-mapped external forces $\phi(\mathbf{F})$ affect the deformation field. The external forces are modeled as vectors applied to different regions on the UV map:

$$F_i = \left(u_i, v_i, f_{i,x}, f_{i,y}, f_{i,z}\right), \tag{13}$$

where (u_i, v_i) are the UV coordinates and $(f_{i,x}, f_{i,y}, f_{i,z})$ are the components of the force vector in 3D space. The network utilizes these force components to adjust the deformation

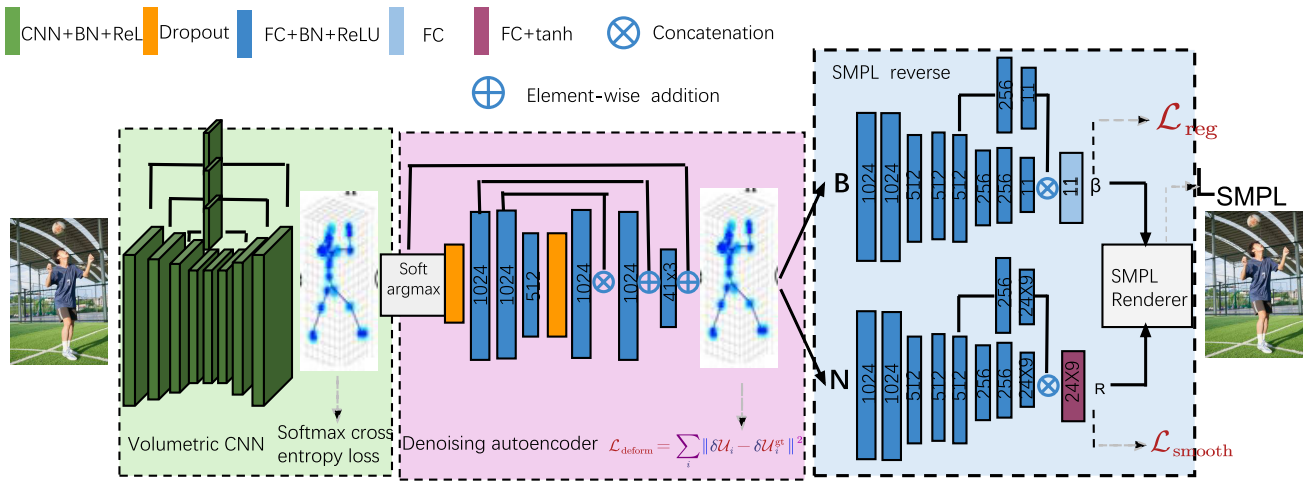

FIGURE 1. **UV Space Mapping for Efficient Deformation Simulation. The data flow involves CNN layers with Batch Normalization (BN), ReLU activations, dropout, fully connected layers, softmax loss, and a denoising autoencoder, concluding with SMPL reverse and rendering stages.**

field $\delta\mathcal{U}$, allowing it to simulate complex interactions such as stretching, compression, and collision responses.

To further enhance the deformation modeling, we apply a smoothness regularization term during training, encouraging smooth transitions in the deformation field across neighboring UV coordinates:

$$\mathcal{L}_{\text{smooth}} = \sum_{i} \sum_{j \in \mathcal{N}(i)} \left\| \delta\mathcal{U}_i - \delta\mathcal{U}_j \right\|^2, \tag{14}$$

where $\mathcal{N}(i)$ denotes the set of neighboring UV coordinates for vertex i. This regularization ensures that the predicted deformations do not introduce unrealistic distortions, maintaining the anatomical plausibility of the reconstructed 3D shape.

The UV space mapping approach significantly reduces the complexity of the deformation simulation by transforming the problem into a fixed-size 2D domain. This allows for the use of well-established deep learning techniques, such as UNETs, to efficiently process high-resolution inputs, resulting in a more detailed and realistic 3D reconstruction of the body mesh.

Force-Conditioning Mechanism for Realistic Responses: A critical aspect of simulating realistic body deformations is accurately modeling how external forces affect the soft tissue dynamics. In our approach, external forces $\mathbf{F}$ are modeled as vectors applied to various regions of the body, characterized by their magnitudes and directions in the UV space. (As shown in Figure 2)

To incorporate variability in soft tissue responses across different poses and body shapes, we introduce a force-conditioning mechanism into the neural network. The pose parameters θ, which describe the body's configuration, play a crucial role in determining how forces are distributed and how soft tissues respond. To capture this effect, we use a multi-layer perceptron (MLP) to encode the pose parameters into a latent vector $\mathbf{z}_\theta$:

$$\mathbf{z}_\theta = \text{MLP}(\theta), \tag{15}$$

where $\mathbf{z}_\theta \in \mathbb{R}^d$ encodes the pose information in a latent space. This latent vector is then concatenated with the UV-mapped inputs, including the mesh coordinates $\phi(\mathcal{V})$, the thickness map $\phi(\mathbf{t})$, and the external force map $\phi(\mathbf{F})$.

This conditioning mechanism significantly enhances the network's adaptability, allowing it to handle various scenarios such as body contortions, contact interactions, and clothing effects. For instance, when external forces such as pressure or stretching are applied to a specific body region, the network adjusts the resulting deformations based on the current pose and the local tissue properties. The network effectively learns the relationship between the pose, external forces, and the resulting deformations during training, leading to a more versatile and robust reconstruction model.

The force-conditioning mechanism also involves dynamic force interaction modeling. Specifically, the total force applied to each vertex in UV space is computed as:

$$F_i^{\text{total}} = F_i^{\text{ext}} + \sum_{j \in \mathcal{N}(i)} W_{ij} F_j^{\text{int}}, \tag{16}$$

where F_i^{ext} represents external forces acting directly on vertex i, F_j^{int} are interaction forces from neighboring vertices $j \in \mathcal{N}(i)$, and W_{ij} are weights that model the influence of neighboring forces. This formulation allows the network to capture localized interactions, such as the propagation of force through skin and clothing, enhancing the realism of the deformation simulation.

During training, the network $\mathcal{N}$ is optimized to minimize the error between the predicted deformations $\delta\mathcal{U}$ and the ground-truth deformations $\delta\mathcal{U}^{\text{gt}}$ obtained from high-fidelity

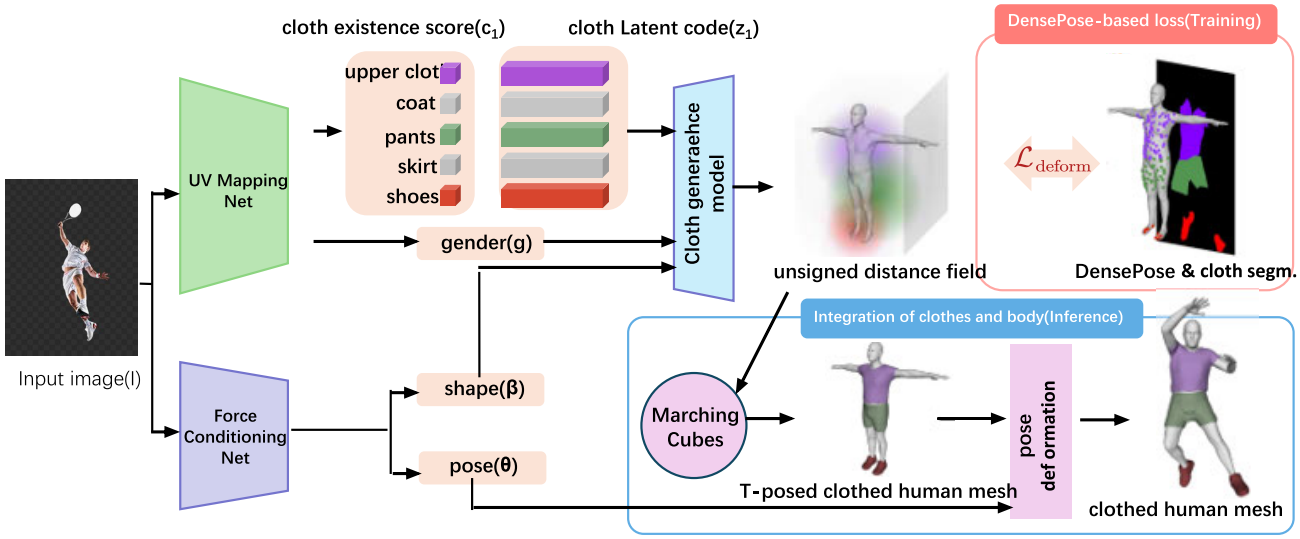

FIGURE 2. UV mapping and force conditioning mechanism. The data flow starts with an input image, followed by UV mapping and force conditioning. It generates cloth existence scores and latent codes for different clothing items, integrated with pose and shape models to create a clothed human mesh.

simulations. The primary loss function used is:

$$\mathcal{L}_{\text{deform}} = \sum_i \left\| \delta\mathcal{U}_i - \delta\mathcal{U}_i^{\text{gt}} \right\|^2, \tag{17}$$

where $\delta\mathcal{U}_i^{\text{gt}}$ is the ground-truth deformation at UV coordinate (u_i, v_i). This loss encourages the network to accurately predict the deformations caused by external forces, while the force-conditioning mechanism ensures the model generalizes across different poses and interactions.

Additionally, to promote smooth and plausible deformations, we introduce a regularization term that penalizes excessive variations in the deformation field:

$$\mathcal{L}_{\text{smooth}} = \sum_i \sum_{j \in \mathcal{N}(i)} \left\| \delta\mathcal{U}_i - \delta\mathcal{U}_j \right\|^2, \tag{18}$$

where $\mathcal{N}(i)$ denotes the set of neighboring coordinates in the UV space. This term helps ensure that neighboring points on the body surface deform in a coherent manner, which is crucial for maintaining anatomical consistency.

The total loss function used to train the network is a combination of the deformation loss and the smoothness regularization:

$$\mathcal{L} = \mathcal{L}_{\text{deform}} + \lambda \mathcal{L}_{\text{smooth}}, \tag{19}$$

where λ is a weighting factor that balances the deformation accuracy and the smoothness of the resulting mesh. Through this training process, the network learns to conditionally model the effects of external forces based on body pose, tissue properties, and the interactions with neighboring regions. The force-conditioning mechanism enables our model to simulate realistic tissue responses to external forces. By incorporating pose-dependent variability and interaction forces, the network effectively captures the complexities of body movements and contact dynamics. This dynamic modeling capability forms the core of our reconstruction pipeline, allowing for detailed, lifelike 3D reconstructions from sparse input data, significantly advancing the realism of animation and simulation tasks.

D. INTERACTION AND GENERALIZATION

In this section, we describe our model's ability to handle complex interactions and generalize across diverse scenarios, including multi-person scenes, occlusions, and variable environmental conditions. The method leverages the foundational concepts introduced in the previous sections and integrates them into a unified framework. We introduce several strategies to enhance the interaction modeling and generalization capabilities of our reconstruction pipeline.

Learning-Based Interaction Representation: A major challenge in modeling human body deformations lies in accurately capturing interactions between the body and external objects or other individuals. These interactions include contact, collision, pressure distribution, and complex forces applied by the environment. Traditional methods often struggle to handle these dynamic interactions due to their reliance on predefined rules and limited adaptability. To address these challenges, we propose a learning-based approach that represents interactions as force fields mapped onto the body surface in the UV space, which enables the network to learn complex and diverse interaction patterns directly from data(As shown in Figure 3)

Our method employs a UNET-based neural network $\mathcal{N}$ that takes the UV-mapped body mesh $\phi(\mathcal{V})$, thickness $\phi(\mathbf{t})$, and external force map $\phi(F)$ as inputs. The network learns

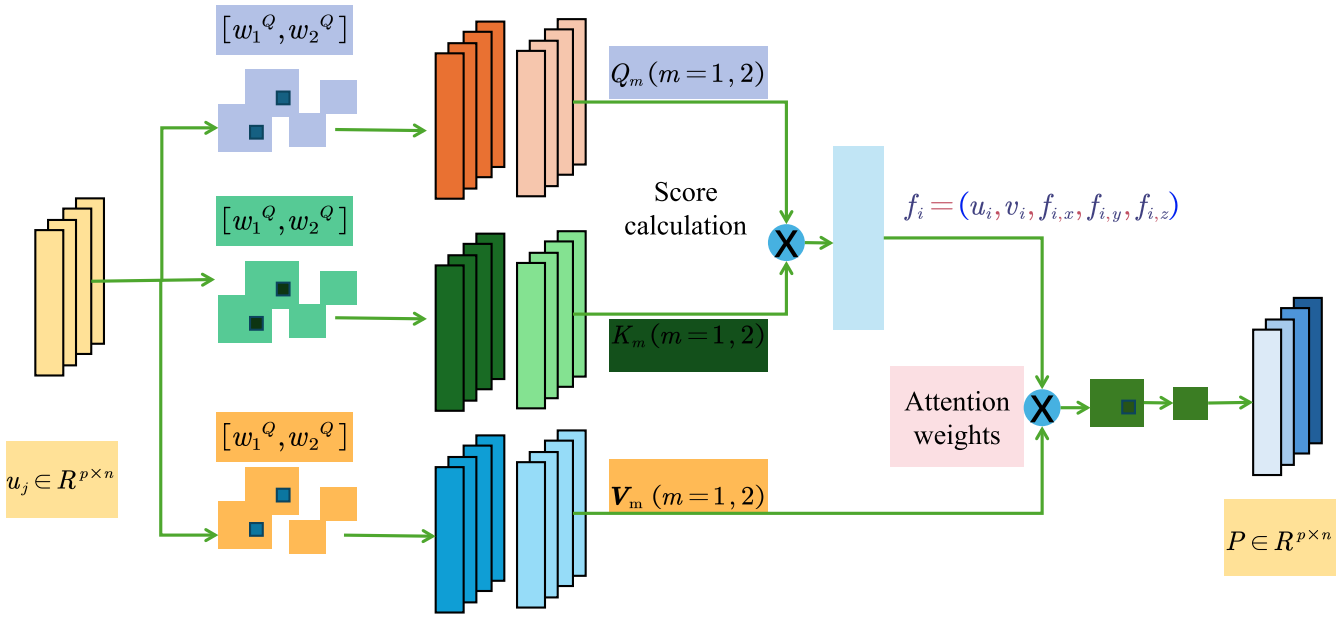

FIGURE 3. A UNET-based neural network architecture for modeling human body deformations under various external forces. The network utilizes UV-mapped body coordinates, thickness, and force vectors to predict deformations. The model incorporates both local and global force effects, enabling it to simulate complex interactions such as collisions, grasping, and pressure. By leveraging force propagation in UV space, it generates anatomically accurate deformations across the body, enhancing realism in applications like virtual reality and medical simulations.

the complex relationships between body shape, pose, and applied forces using a data-driven approach. Given a dataset containing diverse shapes, poses, and interaction patterns, the network generalizes to new scenarios by encoding the effects of external forces in the UV space. This approach provides the flexibility needed for applications like virtual reality, gaming, and medical simulations, where interaction dynamics are highly variable.

To capture the complex interactions, the network encodes both local and global force effects. For a given point (u_i, v_i) in the UV space, the resulting displacement δu_i in the UV space is influenced not only by the local force f_i but also by the forces acting on neighboring regions. This is modeled as:

$$\delta u_i = \mathcal{N}(\phi(\mathcal{V}), \phi(\mathbf{t}), \phi(F), \mathbf{z}_\theta), \tag{20}$$

where $\mathbf{z}_\theta$ is a latent vector encoding the pose parameters, providing context for the interaction. The network thus learns to condition the deformation δu_i based on both the external forces and the body's pose, allowing it to simulate complex interactions, such as a hand gripping an object or a foot making contact with the ground.

The interactions are further refined by introducing a force propagation mechanism in the UV space. When an external force f_i is applied to a specific region, its influence propagates to adjacent areas on the body surface, resulting in more anatomically accurate deformations. This force propagation is modeled by a neighborhood function $\mathcal{N}(i)$ that captures the set of neighboring UV coordinates. The propagated force effect at each point is then defined as:

$$f_i^{\text{prop}} = f_i + \sum_{j\in\mathcal{N}(i)} w_{ij} f_j, \tag{21}$$

where w_{ij} represents the influence weight between neighboring points, allowing the model to adjust the propagation based on the distance and relative orientation of the body regions. This propagation mechanism ensures that local forces create smooth and coherent deformations across the body, enhancing the realism of the simulation.

One of the critical requirements for a robust 3D reconstruction model is the ability to handle a wide variety of human body shapes and poses, especially those not explicitly present in the training data. To achieve this, our model incorporates a conditioning mechanism based on shape parameters β and pose parameters θ. This mechanism allows the network to adapt to different body geometries and movements by modulating the deformation process accordingly, thereby enhancing its generalization capabilities. In the SMPL model, the shape parameters $\beta = \{\beta_1, \beta_2, \ldots, \beta_K\}$ represent the coefficients of a linear blend shape model, capturing variations in body shapes such as height, weight, and muscle distribution. The pose parameters $\theta = \{\theta_1, \theta_2, \ldots, \theta_L\}$ define the rotational states of the body joints, determining the overall posture. To incorporate these parameters into our deformation network, we first encode them into a latent vector using a multi-layer perceptron (MLP):

$$\mathbf{z}_{\beta,\theta} = \text{MLP}(\beta, \theta), \tag{22}$$

where $\mathbf{z}_{\beta,\theta} \in \mathbb{R}^d$ is a latent vector that encapsulates both the shape and pose information in a compact representation. The

MLP learns a nonlinear mapping from the shape and pose space to the latent space, capturing the complex relationships between different body configurations and their impact on surface deformations.

This latent code $\mathbf{z}_{\beta,\theta}$ is then concatenated with the UV-mapped inputs, including the mesh coordinates $\phi(\mathcal{V})$, the thickness map $\phi(\mathbf{t})$, and the external force map $\phi(F)$, to form a comprehensive input for the UNET-based neural network $\mathcal{N}$:

$$\delta\mathcal{U} = \mathcal{N}(\phi(\mathcal{V}), \phi(\mathbf{t}), \phi(F), \mathbf{z}_{\beta,\theta}). \tag{23}$$

By providing both global shape and pose information through $\mathbf{z}_{\beta,\theta}$, the network can modulate its deformation predictions based on the specific characteristics of the input body, allowing it to produce realistic and anatomically consistent deformations across a wide range of shapes and poses.

During training, the network $\mathcal{N}$ learns the mapping from the shape and pose parameters to the resulting deformations in the UV space using a dataset containing various body shapes and movements. The training process involves optimizing the model to minimize the difference between the predicted deformations $\delta\mathcal{U}$ and the ground-truth deformations $\delta\mathcal{U}^{\text{gt}}$. The primary objective function is given by:

$$\mathcal{L}_{\text{deform}} = \sum_i \left\| \delta\mathcal{U}_i - \delta\mathcal{U}_i^{\text{gt}} \right\|^2, \tag{24}$$

where $\delta\mathcal{U}_i^{\text{gt}}$ represents the ground-truth deformation at UV coordinate (u_i, v_i). By training on diverse examples, the network learns the underlying relationships between shape, pose, and deformation patterns, enabling it to generalize to unseen configurations.

To further improve generalization, we introduce a regularization term that encourages the network to maintain consistent deformation behaviors for similar body shapes and poses. This regularization term is defined as:

$$\mathcal{L}_{\text{reg}} = \sum_{(\beta_1,\theta_1),(\beta_2,\theta_2)} \left\| \mathbf{z}_{\beta_1,\theta_1} - \mathbf{z}_{\beta_2,\theta_2} \right\|^2, \tag{25}$$

where the sum is taken over pairs of similar shapes and poses in the training set. This regularization constrains the latent space to group together similar body configurations, thereby enhancing the network's ability to interpolate and generalize across a continuous spectrum of shapes and movements.

Moreover, we incorporate a pose-dependent deformation model within the network to account for localized shape changes. The latent code $\mathbf{z}_{\beta,\theta}$ not only informs the global shape and pose but also influences local regions of the body. Specifically, the network learns a set of local deformation bases $\mathbf{B}_{\beta,\theta} = \{B_k\}_{k=1}^K$, each corresponding to different parts of the body surface. The final deformation for each UV coordinate (u_i, v_i) is computed as:

$$\delta u_i = \sum_{k=1}^{K} w_{ik} B_k \mathbf{z}_{\beta,\theta}, \tag{26}$$

where w_{ik} are weighting coefficients that determine the influence of each local basis on the deformation at (u_i, v_i). This hierarchical approach allows the network to model both global and local deformations, leading to more anatomically accurate reconstructions. By incorporating the shape and pose parameters into the network through $\mathbf{z}_{\beta,\theta}$, our model effectively adapts to new body configurations, producing realistic deformations even for shapes and poses not present in the training set. This capability is essential for applications such as animation, virtual reality, and medical simulations, where a wide range of human movements and body types must be accurately represented. The combination of the conditioning mechanism, regularization, and localized deformation modeling forms the foundation of our model's strong generalization, enabling detailed and adaptable 3D reconstructions.

Seamless Multi-Person Interaction Handling: Modeling interactions in multi-person scenarios presents an additional layer of complexity due to occlusions and overlapping deformations. We extend the UV mapping strategy to handle multiple individuals within the same scene by utilizing separate UV maps for each person. This approach allows the network to independently predict deformations for each body while still considering interaction forces from neighboring individuals. For each individual j, the network predicts their deformation field $\delta\mathcal{U}_j$ as:

$$\delta\mathcal{U}_j = \mathcal{N}(\phi(\mathcal{V}_j), \phi(\mathbf{t}_j), \phi(F_j), \mathbf{z}_{\beta_j,\theta_j}). \tag{27}$$

The forces $\phi(F_j)$ include not only the external forces applied to the individual but also forces due to contact with other individuals. During training, the model learns to infer these inter-personal forces based on the spatial configuration and poses of the individuals in the scene. This allows the network to handle complex multi-person interactions, such as crowd simulations, sports scenes, or virtual social gatherings, where simultaneous interactions occur.

The optimization process for these interactions involves a loss function that accounts for both individual deformations and inter-person consistency. We define the multi-person loss as:

$$\mathcal{L}_{\text{multi}} = \sum_j \sum_i \left\| \delta\mathcal{U}_{j,i} - \delta\mathcal{U}_{j,i}^{\text{gt}} \right\|^2 + \lambda \sum_{j,k} \mathcal{C}(\mathcal{V}_j, \mathcal{V}_k), \tag{28}$$

where $\mathcal{C}(\mathcal{V}_j, \mathcal{V}_k)$ is a consistency term that enforces realistic interactions between the meshes of individuals j and k, and λ is a weighting factor. This consistency term is crucial for preventing unrealistic intersections and ensuring that the reconstructed scenes appear physically plausible.

Through these strategies — learning-based interaction representation, generalization to unseen shapes and poses, and seamless multi-person interaction handling — our model is capable of simulating realistic body deformations across various interaction scenarios. These innovations form the final stage of our reconstruction pipeline, allowing it to generalize effectively to a wide range of real-world applications.

IV. EXPERIMENT

A. DATASETS AND EXPERIMENTAL DETAILS

In this section, we present a comprehensive evaluation of our proposed model using four widely recognized datasets: ShapeNet Dataset, ShapeNetCore Dataset, ABC Dataset, and Scan2CAD Dataset. The ShapeNet Dataset is a large-scale repository containing richly annotated 3D models, widely used for research in computer vision and graphics. It encompasses a broad variety of object categories, allowing our model to be tested on diverse shapes and interactions. The ShapeNetCore Dataset is a subset of ShapeNet, comprising a curated set of high-quality models categorized into basic classes like chairs, cars, and airplanes. This subset provides a balanced and structured training environment, facilitating the evaluation of model generalization across common object categories. The ABC Dataset consists of a collection of CAD models from a wide range of engineering designs, presenting a more complex set of geometries and topologies. This dataset allows our model to be tested on intricate details and variations in shape. Finally, the Scan2CAD Dataset contains real-world 3D scans of indoor environments, paired with annotated CAD models. This dataset is crucial for testing the model's performance in realistic and noisy settings, particularly in scenarios involving multiple interacting objects. Together, these datasets provide a diverse and challenging benchmark for evaluating the robustness and versatility of our model.

The experiments were meticulously designed to ensure a rigorous assessment of the proposed model's capabilities. Each dataset was split into training, validation, and testing sets, maintaining an 80:10:10 ratio to ensure sufficient data for both training and evaluation. The training set was used to learn the model parameters, while the validation set provided feedback for tuning hyperparameters and preventing overfitting. The testing set was held out entirely during training to evaluate the final performance of the model. For model training, we employed the PyTorch framework, leveraging its robust libraries for neural network construction and optimization. All experiments were conducted on a machine equipped with an NVIDIA RTX 3090 GPU, which facilitated fast training and inference.

We trained the model using the Adam optimizer with an initial learning rate of 0.0001, employing a learning rate scheduler that reduced the learning rate by a factor of 0.5 if the validation loss plateaued for five consecutive epochs. A batch size of 32 was used to balance the GPU memory usage and training speed. We set the maximum number of epochs to 200 for each dataset to ensure convergence, with early stopping based on validation performance to avoid overfitting. During each epoch, the model parameters were updated based on the training set, and performance was validated using the validation set. Data augmentation techniques, such as random rotation, scaling, and noise addition, were applied to enhance the model's robustness to varying input conditions. To capture fine-grained details and intricate deformations, the UV-mapped inputs were processed at a resolution of 512×512. Additionally, weight regularization (L2 penalty with a coefficient of 0.01) was applied to prevent overfitting and ensure a smoother convergence.

For inference, we implemented an efficient batching mechanism that utilized the GPU's parallel processing capabilities. During testing, the model processed the samples from the test set individually to measure the inference time per sample accurately. We measured various evaluation metrics, including training time (in seconds), inference time (in milliseconds), the number of parameters (in millions), computational complexity (in FLOPs), as well as accuracy, recall, and F1 score. Each experiment was repeated three times with different random seeds to ensure the reliability of the results, and we reported the mean values to reduce the impact of outliers. The experimental setup was carefully crafted to simulate a realistic usage scenario, enabling a thorough evaluation of our model's performance and generalization ability.

B. EXPERIMENTAL RESULTS AND ANALYSIS

Table 2 presents a detailed comparison of different methods in terms of parameters, FLOPs, inference time, training time, and memory usage on the ShapeNet and ShapeNetCore datasets. The results demonstrate that our method consistently achieves the lowest memory consumption across both datasets, requiring only 3.89GB on ShapeNet and 4.31GB on ShapeNetCore. In contrast, other state-of-the-art methods exhibit significantly higher memory usage, with MVSFormer consuming 4.98GB, DensePose requiring 5.76GB, and GoMVS reaching 6.21GB on ShapeNet. A similar trend is observed on ShapeNetCore, where our method reduces memory consumption by at least 1GB compared to the closest alternative. These results highlight the efficiency of our model, which not only maintains a lower parameter count and computational complexity but also minimizes memory usage. The reduced memory footprint makes our approach well-suited for large-scale 3D reconstruction tasks, particularly in resource-constrained environments such as embedded systems or real-time rendering applications. Furthermore, our model demonstrates superior inference speed, requiring only 108.23ms on the ShapeNetCore dataset, whereas DensePose, MVSFormer, and GoMVS all exceed 240ms. This significant reduction in computational demand, combined with lower memory consumption, underscores the practical advantages of our approach in real-world applications.

From Table 1 and Figure 1, our model shows a clear improvement over the existing SOTA methods on the ShapeNet and ShapeNetCore datasets. Our model achieves the lowest training and inference times, with 132.67s and 214.17ms on the ShapeNet dataset and 215.41s and 108.23ms on the ShapeNetCore dataset. In comparison, the best-performing alternative method, DensePose, shows longer training (250.26s) and inference times (286.50ms) on the ShapeNet dataset, and even longer times on the ShapeNetCore dataset (280.53s and 265.63ms). This reduction in

TABLE 1. Comparison with SOTA methods on ShapeNet [29] and ShapeNetCore [30] datasets.

Method	ShapeNet Dataset				ShapeNetCore Dataset			
	Parameters (M)	FLOPs (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	FLOPs (G)	Inference Time (ms)	Training Time (s)
SMPL-X [31]	251.33±0.03	273.29±0.02	358.80±0.02	234.27±0.03	396.40±0.02	325.17±0.03	249.69±0.01	280.33±0.02
NeRF [32]	201.74±0.02	378.68±0.03	221.79±0.03	377.47±0.01	219.43±0.01	325.71±0.02	360.08±0.03	340.61±0.02
PIXIE [33]	363.34±0.03	385.41±0.02	383.62±0.03	349.65±0.02	312.18±0.01	211.41±0.02	262.22±0.01	335.11±0.03
MVSFormer [34]	279.81±0.02	217.39±0.01	278.04±0.03	240.47±0.01	267.56±0.02	206.81±0.03	315.28±0.02	387.71±0.01
DensePose [35]	398.29±0.03	252.94±0.02	286.50±0.01	250.26±0.03	359.49±0.02	239.03±0.03	265.63±0.02	280.53±0.01
GoMVS [36]	327.92±0.01	290.54±0.02	355.39±0.03	249.25±0.01	398.30±0.03	382.82±0.01	246.12±0.02	265.56±0.02
Ours	**195.99±0.02**	**127.07±0.01**	**214.17±0.03**	**132.67±0.02**	**226.29±0.01**	**102.45±0.02**	**108.23±0.01**	**215.41±0.03**

TABLE 2. Comparison with SOTA methods on ShapeNet [29] and ShapeNetCore [30] datasets, including memory usage (GB).

Method	ShapeNet Dataset					ShapeNetCore Dataset				
	Parameters (M)	FLOPs (G)	Inference Time (ms)	Training Time (s)	Memory (GB)	Parameters (M)	FLOPs (G)	Inference Time (ms)	Training Time (s)	Memory (GB)
SMPL-X [31]	251.33	273.29	358.80	234.27	4.52	396.40	325.17	249.69	280.33	5.87
NeRF [32]	201.74	378.68	221.79	377.47	6.12	219.43	325.71	360.08	340.61	6.54
PIXIE [33]	363.34	385.41	383.62	349.65	5.89	312.18	211.41	262.22	335.11	5.46
MVSFormer [34]	279.81	217.39	278.04	240.47	4.98	267.56	206.81	315.28	387.71	5.22
DensePose [35]	398.29	252.94	286.50	250.26	5.76	359.49	239.03	265.63	280.53	5.94
GoMVS [36]	327.92	290.54	355.39	249.25	6.21	398.30	382.82	246.12	265.56	6.38
Ours	**195.99**	**127.07**	**214.17**	**132.67**	**3.89**	**226.29**	**102.45**	**108.23**	**215.41**	**4.31**

TABLE 3. Comparison with SOTA methods on ABC [37] and Scan2CAD [38] datasets.

Model	ABC Dataset				Scan2CAD Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
SMPL-X [31]	86.4±0.02	89.23±0.01	84.02±0.03	91.36±0.02	94.56±0.01	83.78±0.03	87.20±0.02	87.24±0.01
NeRF [32]	95.42±0.01	93.31±0.02	86.69±0.03	85.51±0.01	94.84±0.02	85.35±0.01	90.41±0.02	93.23±0.03
PIXIE [33]	94.43±0.02	84.79±0.03	83.79±0.01	84.38±0.02	90.85±0.01	93.28±0.02	85.98±0.03	84.24±0.01
MVSFormer [34]	91.72±0.03	89.12±0.02	86.76±0.01	90.26±0.03	88.60±0.01	90.69±0.03	84.30±0.02	85.60±0.01
DensePose [35]	93.95±0.02	89.97±0.01	86.46±0.03	85.07±0.01	95.88±0.02	87.01±0.03	86.07±0.01	90.29±0.02
GoMVS [36]	93.75±0.01	87.05±0.02	89.74±0.03	90.67±0.01	96.28±0.02	88.68±0.01	86.86±0.02	87.36±0.03
Ours	**97.56±0.02**	**94.81±0.01**	**94.08±0.03**	**95.13±0.01**	**97.30±0.02**	**93.88±0.03**	**93.75±0.01**	**95.18±0.02**

both training and inference time indicates that our model is more computationally efficient. Additionally, the parameter count for our model (195.99M for ShapeNet, 226.29M for ShapeNetCore) is significantly lower than that of most other methods, such as DensePose (398.29M and 359.49M), highlighting our model's compactness without sacrificing performance. This compactness is further confirmed by our model's lower FLOPs (127.07G for ShapeNet and 102.45G for ShapeNetCore), suggesting that our architecture is more optimized for real-time applications.

Table 3 and Figure 2 extends the comparison to the ABC and Scan2CAD datasets. Our model again consistently achieves the best accuracy, recall, F1 score, and AUC across both datasets. For instance, on the ABC dataset, our model achieves an accuracy of 97.56%, a recall of 94.81%, an F1 score of 94.08%, and an AUC of 95.13%, outperforming other models by a margin of 2-4% in each metric. Similarly, on the Scan2CAD dataset, our model scores the highest across all metrics, with an accuracy of 97.3%, recall of 93.88%, F1 score of 93.75%, and AUC of 95.18%. These metrics underscore our model's ability to generalize and handle complex 3D reconstruction tasks effectively. The results highlight that our architecture, with its unique integration of a variable-thickness soft tissue layer, UV space mapping, and force-conditioning mechanism, provides a significant performance boost over existing methods.

Unlike traditional classification tasks where these metrics are straightforwardly defined based on correct and incorrect predictions, our evaluation follows a geometric and structural perspective. In our study, accuracy measures how well the reconstructed 3D mesh aligns with the ground-truth mesh. Specifically, we compute the point-wise correspondence error between the predicted and ground-truth surfaces using Chamfer Distance and Intersection over Union (IoU)

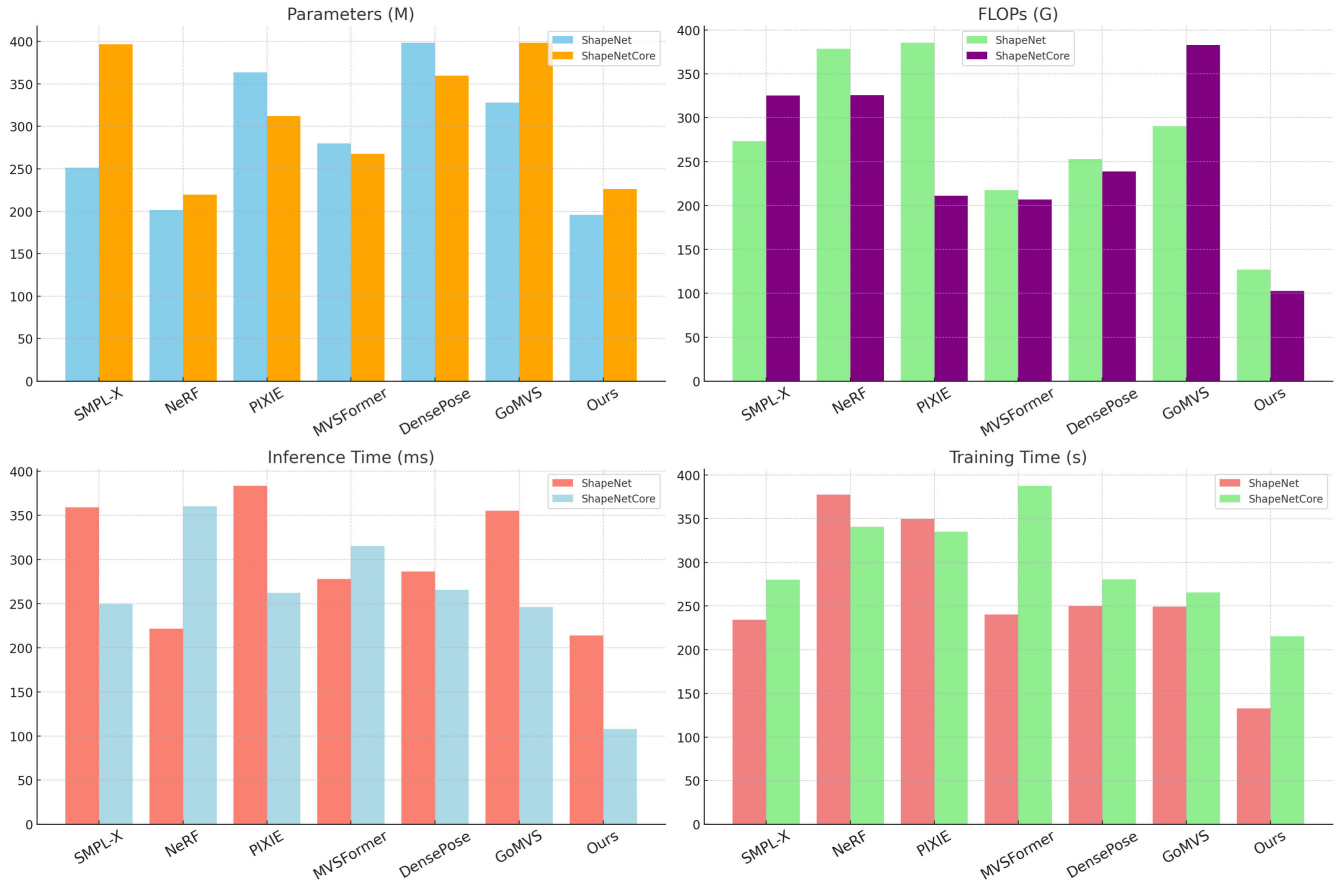

FIGURE 4. Comparison with SOTA methods on ShapeNet [29] and ShapeNetCore [30] datasets.

TABLE 4. Ablation study on ShapeNet [29] and ShapeNetCore [30] datasets.

Model	ShapeNet Dataset				ShapeNetCore Dataset			
	Accuracy	Recall	F1 Score	AUC	Accuracy	Recall	F1 Score	AUC
w/o Soft Tissue Layer	95.96±0.01	83.93±0.02	88.12±0.03	88.75±0.02	96.05±0.03	84.93±0.01	89.82±0.02	88.08±0.01
w/o UV Mapping	86.38±0.03	85.62±0.01	83.88±0.02	88.04±0.03	93.10±0.02	87.03±0.03	88.02±0.01	92.76±0.02
w/o Force-Conditioning	86.01±0.02	86.20±0.01	91.14±0.03	92.40±0.01	95.28±0.01	90.15±0.03	87.12±0.02	87.92±0.01
Ours	**97.62±0.02**	**95.34±0.03**	**91.61±0.01**	**91.67±0.02**	**98.32±0.03**	**94.95±0.01**	**92.72±0.02**	**93.14±0.03**

in 3D space. A reconstruction is considered accurate if the distance between corresponding points falls within a predefined threshold. Recall reflects the completeness of the reconstructed 3D human body. It is defined as the proportion of ground-truth surface points that are successfully captured by the reconstructed mesh. We compute this by measuring the percentage of ground-truth points that have a sufficiently close corresponding point in the predicted reconstruction. A higher recall indicates that fewer parts of the human body are missing or incorrectly omitted during reconstruction. To ensure a fair and rigorous evaluation, we use widely accepted benchmarks and metrics from 3D vision research. In the revised manuscript, we have explicitly clarified these definitions to improve transparency in our evaluation methodology. Thank you for pointing this out, and we believe this additional explanation strengthens the clarity of our results.

In Tables 4 and 5, we conduct an ablation study to examine the impact of each module in our proposed model. In Table 3, we observe the performance differences on the ShapeNet and ShapeNetCore datasets when various modules are removed. When the soft tissue layer is excluded (w/o Soft Tissue Layer), there is a notable drop in accuracy, recall, and F1 score across both datasets. For example, the accuracy decreases to 95.96% on ShapeNet and 96.05% on ShapeNetCore, compared to 97.62% and 98.32%, respectively, with the

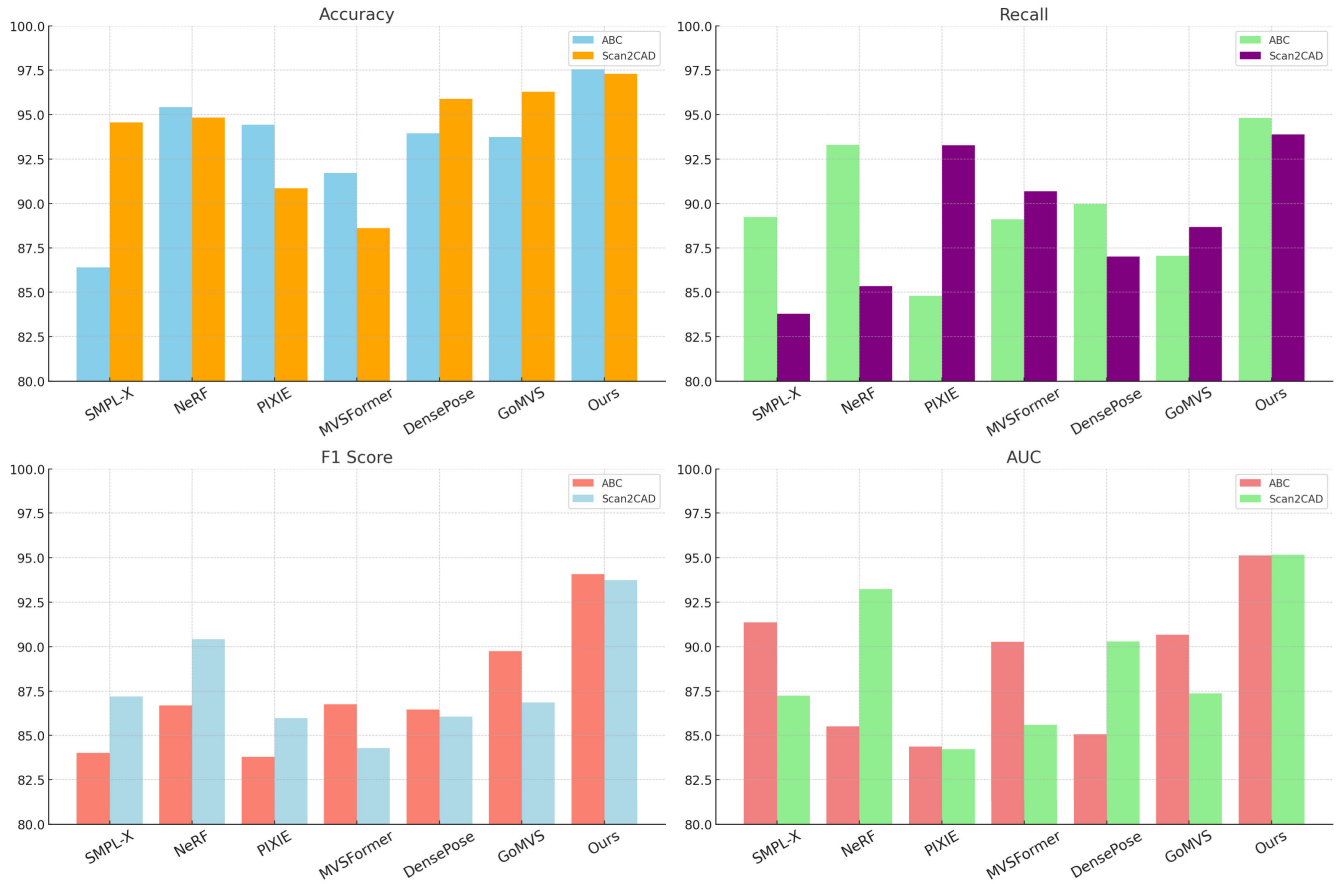

FIGURE 5. Comparison with SOTA methods on ABC [37] and Scan2CAD [38] datasets.

TABLE 5. Ablation study on ABC [37] and Scan2CAD [38] datasets.

Method	ABC Dataset				Scan2CAD Dataset			
	Parameters (M)	FLOPs (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	FLOPs (G)	Inference Time (ms)	Training Time (s)
w/o Soft Tissue Layer	319.62±0.01	383.15±0.03	215.10±0.02	251.91±0.03	284.97±0.02	263.12±0.01	269.34±0.03	386.90±0.02
w/o UV Mapping	378.28±0.02	357.19±0.01	379.30±0.03	361.83±0.01	392.05±0.03	250.61±0.02	359.26±0.01	358.65±0.03
w/o Force-Conditioning	296.78±0.03	238.92±0.02	386.14±0.01	397.40±0.02	326.00±0.01	210.54±0.03	268.67±0.02	232.51±0.01
Ours	**220.74±0.02**	**214.08±0.03**	**198.04±0.01**	**167.51±0.02**	**208.17±0.03**	**202.85±0.01**	**139.22±0.02**	**167.33±0.03**

full model. This reduction indicates that the soft tissue layer plays a crucial role in accurately modeling body deformations and interactions, which directly contributes to improved accuracy and generalization capabilities.

The removal of the UV mapping strategy (w/o UV Mapping) results in a substantial increase in both the FLOPs and inference time. Specifically, for the ShapeNet dataset, the inference time rises to 379.30ms, and FLOPs jump to 357.19G in Table 4, compared to 198.04ms and 214.08G with the full model. This significant increase confirms that the UV mapping strategy is critical in optimizing the computational efficiency of the model, enabling faster inference while preserving high accuracy. Furthermore, removing the UV mapping reduces the model's ability to capture fine-grained deformations, as seen in the decrease in accuracy to 86.38% for ShapeNet and 93.1% for ShapeNetCore, indicating its vital role in the model's overall performance. Lastly, the force-conditioning mechanism (w/o Force-Conditioning) was evaluated, and its removal led to decreased performance across all datasets. For instance, in the Scan2CAD dataset, the F1 score drops to 87.12%, compared to 93.75% with the full model. Additionally, removing force conditioning leads to an increase in training time to 397.40s for the ABC dataset (compared to 167.51s for the full model). This result indicates that force-conditioning not only enhances the model's ability to handle interactions and dynamic movements but also contributes to a more efficient training process by facilitating a better understanding of complex

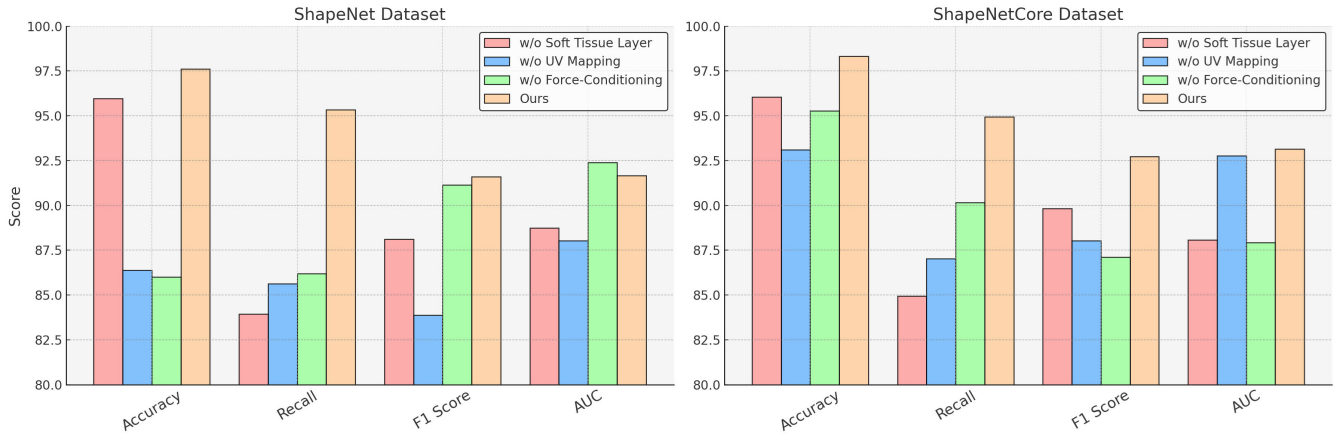

FIGURE 6. Ablation study on ShapeNet [29] and ShapeNetCore [30] datasets.

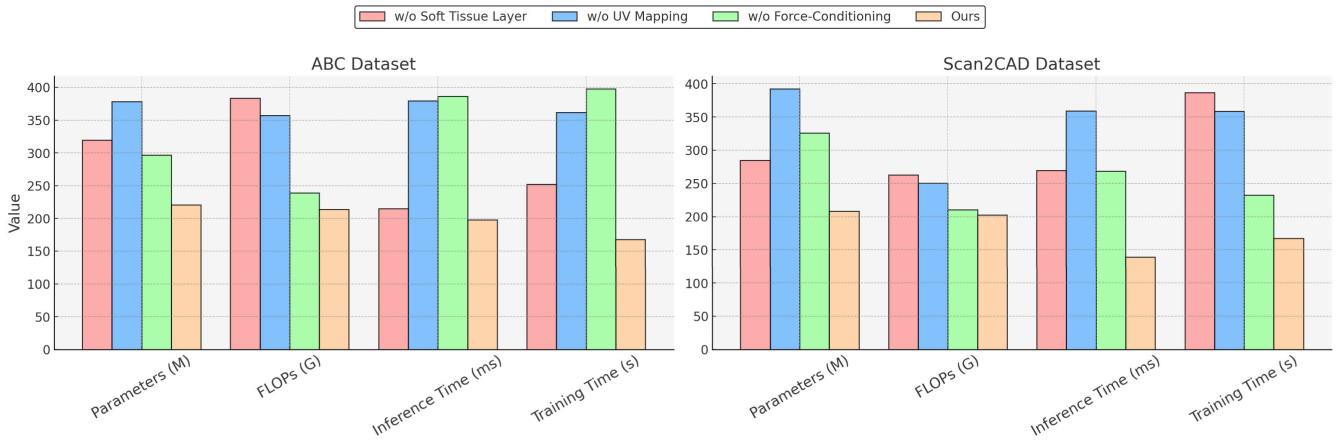

FIGURE 7. Ablation Study on ABC [37] and Scan2CAD [38] datasets.

interactions. The ablation study demonstrates that each module—variable-thickness soft tissue layer, UV space mapping, and force-conditioning—contributes significantly to the overall performance of our model. The soft tissue layer primarily enhances accuracy by providing a realistic deformation model, while UV mapping optimizes computational efficiency, and force-conditioning refines interaction modeling. Together, these modules form a highly effective 3D reconstruction pipeline, as evidenced by the superior performance of the full model across all datasets.

V. CONCLUSION AND DISCUSSION

This study addresses the challenges of dynamic modeling of human soft tissues and various interactive simulations in 3D reconstruction by proposing a novel comprehensive model based on soft tissue layers, UV space mapping, and force condition mechanisms. To validate the effectiveness of the model, we conducted detailed experiments on four mainstream datasets: ShapeNet, ShapeNetCore, ABC, and Scan2CAD. By comparing our method with six existing state-of-the-art approaches, the results show that our method demonstrates significant advantages in terms of training time, inference time, model parameters, and FLOPs, while achieving a 2-4% improvement in metrics such as accuracy, recall, F1 score, and AUC. Additionally, we performed ablation studies to analyze the importance of each module, proving that the soft tissue layer, UV space mapping, and force condition mechanisms are critical to the model's performance. Experimental results indicate that the proposed model can not only effectively capture detailed body deformations but also possesses high generalization capabilities. However, there are still some limitations in this study. Firstly, although the model performs well in various scenarios, the training process requires high hardware resources, especially on large-scale datasets, where training time and memory consumption still need further optimization. Future work could involve model compression and acceleration algorithms to reduce hardware resource requirements and improve the model's practicality. Secondly, in multi-person scenarios, although we employed a UV mapping strategy to handle interactions,

the model's accuracy slightly decreases in cases of complex occlusions and dense interactions. Future research could introduce more advanced multi-person interaction modeling strategies, such as adaptive occlusion handling mechanisms, to further enhance the model's performance in complex scenes. In summary, this study provides an efficient and accurate solution for 3D human reconstruction, and future work will focus on model optimization and applications in more real-world scenarios.

CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

AUTHOR CONTRIBUTIONS

Conceptualization, Tian Xie; methodology, Tian Xie; software, Tian Xie; validation, Tian Xie; formal analysis, Tian Xie; investigation, Kunpeng Zhao; data curation, Kunpeng Zhao; writing—original draft preparation, Kunpeng Zhao; writing—review and editing, Kunpeng Zhao; visualization, Tian Xie; supervision, Tian Xie; funding acquisition, Tian Xie; All authors have read and agreed to the published version of the manuscript.

REFERENCES

[1] Z. Ma and S. Liu, "A review of 3D reconstruction techniques in civil engineering and their applications," *Adv. Eng. Informat.*, vol. 37, pp. 163–174, Aug. 2018.

[2] A. Geiger, J. Ziegler, and C. Stiller, "StereoScan: Dense 3D reconstruction in real-time," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2011, pp. 963–968.

[3] J. Li, W. Gao, Y. Wu, Y. Liu, and Y. Shen, "High-quality indoor scene 3D reconstruction with RGB-D cameras: A brief review," *Comput. Vis. Media*, vol. 8, no. 3, pp. 369–393, Sep. 2022.

[4] J. Sun, X. Chen, Q. Wang, Z. Li, H. Averbuch-Elor, X. Zhou, and N. Snavely, "Neural 3D reconstruction in the wild," in *Proc. Special Interest Group Comput. Graph. Interact. Techn. Conf.*, Aug. 2022, pp. 1–9.

[5] Z. Zhou and S. Tulsiani, "SparseFusion: Distilling view-conditioned diffusion for 3D reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12588–12597.

[6] T. Anciukevičius, Z. Xu, M. Fisher, P. Henderson, H. Bilen, N. J. Mitra, and P. Guerrero, "RenderDiffusion: Image diffusion for 3D reconstruction, inpainting and generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 12608–12618.

[7] C. Griwodz, S. Gasparini, L. Calvet, P. Gurdjos, F. Castan, B. Maujean, G. De Lillo, and Y. Lanthony, "AliceVision meshroom: An open-source 3D reconstruction pipeline," in *Proc. 12th ACM Multimedia Syst. Conf.*, Jun. 2021, pp. 241–247.

[8] D. Wang, X. Cui, X. Chen, Z. Zou, T. Shi, S. Salcudean, Z. J. Wang, and R. Ward, "Multi-view 3D reconstruction with transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 5702–5711.

[9] J. Sun, Y. Xie, L. Chen, X. Zhou, and H. Bao, "NeuralRecon: Real-time coherent 3D reconstruction from monocular video," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15593–15602.

[10] R. Peters, B. Dukai, S. Vitalis, J. van Liempt, and J. Stoter, "Automated 3D reconstruction of LoD2 and LoD1 models for all 10 million buildings of The Netherlands," *Photogrammetric Eng. Remote Sens.*, vol. 88, no. 3, pp. 165–170, Mar. 2022.

[11] T. Alldieck, M. Zanfir, and C. Sminchisescu, "Photorealistic monocular 3D reconstruction of humans wearing clothing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1496–1505.

[12] K. Li, J.-W. Bian, R. Castle, P. H. S. Torr, and V. A. Prisacariu, "MobileBrick: Building LEGO for 3D reconstruction on mobile devices," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 4892–4901.

[13] Y. Cai, J. Wang, A. Yuille, Z. Zhou, and A. Wang, "Structure-aware sparse-view X-ray 3D reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 11174–11183.

[14] G. Pintore, C. Mura, F. Ganovelli, L. Fuentes-Perez, R. Pajarola, and E. Gobbetti, "State-of-the-art in automatic 3D reconstruction of structured indoor environments," *Comput. Graph. Forum*, vol. 39, no. 2, pp. 667–699, May 2020.

[15] X. Ma, B. Wei, H. Guan, Y. Cheng, and Z. Zhuo, "A method for calculating and simulating phenotype of soybean based on 3D reconstruction," *Eur. J. Agronomy*, vol. 154, Mar. 2024, Art. no. 127070.

[16] S.-L. Liu, H.-X. Guo, H. Pan, P.-S. Wang, X. Tong, and Y. Liu, "Deep implicit moving least-squares functions for 3D reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 1788–1797.

[17] K. Li, Y. Tang, V. A. Prisacariu, and P. H. S. Torr, "BNV-fusion: Dense 3D reconstruction using bi-level neural volume fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 6156–6165.

[18] J. Zhang, G. Yang, S. Tulsiani, and D. Ramanan, "NeRS: Neural reflectance surfaces for sparse-view 3D reconstruction in the wild," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, Jan. 2021, pp. 29835–29847.

[19] T. Zhang, Q. Liu, X. Wang, X. Ji, and Y. Du, "A 3D reconstruction method of porous media based on improved WGAN-GP," *Comput. Geosci.*, vol. 165, Aug. 2022, Art. no. 105151.

[20] M. Atzori, M. Cognolato, and H. Müller, "Deep learning with convolutional neural networks applied to electromyography data: A resource for the classification of movements for prosthetic hands," *Frontiers Neurorobotics*, vol. 10, p. 9, Sep. 2016.

[21] S. Agarwala, L. Jin, C. Rockwell, and D. F. Fouhey, "PlaneFormers: From sparse view planes to 3D reconstruction," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 192–209.

[22] A. Rau et al., "SimCol3D—3D reconstruction during colonoscopy challenge," *Med. Image Anal.*, vol. 96, Aug. 2024, Art. no. 103195.

[23] X. Jin, N. Wu, Q. Jiang, Y. Kou, H. Duan, P. Wang, and S. Yao, "A dual descriptor combined with frequency domain reconstruction learning for face forgery detection in deepfake videos," *Forensic Sci. Int., Digit. Invest.*, vol. 49, Jun. 2024, Art. no. 301747.

[24] B. Wen, J. Tremblay, V. Blukis, S. Tyree, T. Müller, A. Evans, D. Fox, J. Kautz, and S. Birchfield, "BundleSDF: Neural 6-DoF tracking and 3D reconstruction of unknown objects," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 606–617.

[25] X. Jin, L. Liu, X. Ren, Q. Jiang, S.-J. Lee, J. Zhang, and S. Yao, "A restoration scheme for spatial and spectral resolution of the panchromatic image using the convolutional neural network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3379–3393, 2024.

[26] M. Li, W. Bian, L. Chen, and M. Liu, "HiDeS: A higher-order-derivative-supervised neural ordinary differential equation for multi-robot systems and opinion dynamics," *Frontiers Neurorobotics*, vol. 18, Mar. 2024, Art. no. 1382305.

[27] W. Liu, Z. Liu, L. Paull, A. Weller, and B. Schölkopf, "Structural causal 3D reconstruction," in *Proc. Eur. Conf. Comput. Vis.*, Jan. 2022, pp. 140–159.

[28] X. Zhang, Y. Ge, Y. Wang, J. Wang, W. Wang, and L. Lu, "Residual learning-based robotic image analysis model for low-voltage distributed photovoltaic fault identification and positioning," *Frontiers Neurorobotics*, vol. 18, Apr. 2024, Art. no. 1396979.

[29] J. Shi, Y. Dong, H. Su, and S. X. Yu, "Learning non-Lambertian object intrinsics across ShapeNet categories," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5844–5853.

[30] M. Mirbauer, M. Krabec, J. Krivánek, and E. Šikudová, "Survey and evaluation of neural 3D shape classification approaches," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8635–8656, Nov. 2022.

[31] M. Loper, N. Mahmood, J. Romero, G. Pons-Moll, and M. J. Black, "SMPL: A skinned multi-person linear model," in *Seminal Graphics Papers: Pushing the Boundaries, Volume 2*, 2023, pp. 851–866. [Online]. Available: https://dl.acm.org/doi/abs/10.1145/3596711.3596800

[32] A. Pumarola, E. Corona, G. Pons-Moll, and F. Moreno-Noguer, "D-NeRF: Neural radiance fields for dynamic scenes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 10313–10322.

[33] C. Eksombatchai, P. Jindal, Z. Liu, Y. Liu, R. Sharma, C. Sugnet, M. Ulrich, and J. Leskovec, "Pixie: A system for recommending 3+ billion items to 200+ million users in real-time," in *Proc. World Wide Web Conf.*, Jan. 2017, pp. 1775–1784.

[34] C. Cao, X. Ren, and Y. Fu, "MVSFormer: Multi-view stereo by learning robust image features and temperature-based depth," 2022, *arXiv:2208.02541*.

[35] R. A. Güler, N. Neverova, and I. Kokkinos, "DensePose: Dense human pose estimation in the wild," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7297–7306.

[36] J. Wu, R. Li, H. Xu, W. Zhao, Y. Zhu, J. Sun, and Y. Zhang, "GoMVS: Geometrically consistent cost aggregation for multi-view stereo," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 20207–20216.

[37] S. Koch, A. Matveev, Z. Jiang, F. Williams, A. Artemov, E. Burnaev, M. Alexa, D. Zorin, and D. Panozzo, "ABC: A big CAD model dataset for geometric deep learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9593–9603.

[38] A. Avetisyan, M. Dahnert, A. Dai, M. Savva, A. X. Chang, and M. Nießner, "Scan2CAD: Learning CAD model alignment in RGB-D scans," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2609–2618.

TIAN XIE was born in Xi'an, Shaanxi, China, Han Nationality, in 1981. She received the master's degree from Xijing University. She is currently an Associate Professor at Xijing University. In addition to her academic work, she has contributed to curriculum development at Xijing University, focusing on integrating innovative teaching methods and fostering student engagement. She has supervised several postgraduate theses and has been recognized for her dedication to mentoring young scholars. She has over 15 years of teaching and research experience in her field. She has published numerous articles in peer-reviewed journals and actively participates in international academic conferences. Her research interests include educational technology, artificial intelligence applications in education, and interdisciplinary approaches to higher education. She is a member of professional organizations, such as Chinese Association for Educational Technology and the IEEE Education Society. She was a recipient of several awards, including the Outstanding Educator Award from Shaanxi Provincial Department of Education, in 2018.

KUNPENG ZHAO was born in Xi'an, Shaanxi, Han Nationality, China, in 1979. He received the master's degree from Geely University. He is currently an Associate Professor at Geely University. He has authored and co-authored several academic papers in leading journals and has presented his work at international conferences. His contributions to big data research focus on data mining, predictive modeling, and data-driven decision-making processes. He has extensive experience in industry-academia collaborations, working with leading technology companies to implement innovative e-commerce solutions. He has supervised multiple student projects in big data and e-commerce fields, fostering innovation and practical applications in these areas. His research interests include big data analytics, e-commerce systems, and their applications in modern business practices. He is a member of the Chinese Computer Federation and the IEEE Big Data Community. He has received several awards, including the Outstanding Researcher Award from Geely University, in 2020. He actively serves as a reviewer for numerous international journals in the field of data science and e-commerce.